



Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: 2025-11-4	Entry: 1
Description	A ransomware attack was discovered at a small U.S. healthcare clinic that resulted in the encryption of critical patient data and system files. The attack originated from phishing emails that contained malicious attachments, leading to the installation of malware and disruption of business operations.
Tool(s) used	- Email filtering and scanning tools - Antivirus software - Network traffic monitoring tools (e.g., IDS/IPS) - Incident response communication platform
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? An organized group of unethical hackers. ● What happened? Attackers used phishing emails with malicious attachments to install ransomware that encrypted files on multiple systems ● When did the incident occur? Tuesday morning at approximately 9:00 a.m.

	• Where did the incident happen? At a small healthcare clinic in the United States. • Why did the incident happen? The attackers exploited human error through phishing emails and inadequate email security measures.
Additional notes	The clinic should review its email filtering systems, improve employee phishing awareness training, and ensure regular data backups are performed and stored offline. It would also be important to report the breach under HIPAA regulations due to potential patient data exposure. The next step should include containment, eradication, and recovery planning.

Date: 2025-11-4	Entry: 2
Description	Analyzed a packet capture file (.pcap) - Used Wireshark to investigate network traffic associated with a user connecting to a website. The goal was to identify key network details, including source and destination IP addresses, communication protocols, and the type of data exchanged during the connection.
Tool(s) used	- Wireshark
The 5 W's	Capture the 5 W's of an incident. • Who N/A • What N/A • When N/A • Where N/A

	• Why N/A
Additional notes	Used Wireshark filters to isolate TCP and UDP DNS traffic and reviewed packets containing HTTP requests and responses. The source IP represented the internal host, while the destination IP pointed to the public web server. By expanding the protocol layers, I observed handshake sequences, DNS queries, and plaintext data in the payload of TCP packets. This analysis helped confirm how user activity translates into network-level communication and demonstrated the value of filtering techniques in identifying relevant traffic.

Date: 2025-11-4	Entry: 3
Description	Capturing my first packet - Used tcpdump to capture live network packets and observe how data moves between devices on the network. The goal was to understand the structure of packets and identify potential signs of abnormal traffic.
Tool(s) used	- Tcpdump
The 5 W's	Capture the 5 W's of an incident. • Who N/A • What N/A • When N/A • Where N/A • Why N/A
Additional notes	Successfully listed interfaces using tcpdump -D and captured packets with sudo tcpdump -i eth0 -c 10. Applied filters to view only HTTP traffic. The

	session ended by analyzing a provided sample .pcap file to interpret captured data and confirm key details like source and destination IPs. This lab helped reinforce real-world packet capture and filtering techniques for network analysis.
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Date: 2025-11-4	Entry: 4
Description	Investigated a suspicious file hash using VirusTotal, a multi-engine malware analysis platform that scans files and URLs for known threats. The purpose of the investigation was to confirm whether the file was associated with any known malware or indicators of compromise (IoCs).
Tool(s) used	VirusTotal
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? Security analyst within the organization's Security Operations Center (SOC) ● What happened? Used VirusTotal to analyze a suspicious file hash that triggered an alert from internal security systems. ● When did the incident occur? November 9, 2025, during the Detection and Analysis phase of the incident response process. ● Where did the incident happen? Security Operations Center (SOC) monitoring dashboard. ● Why did the incident happen?

	To determine whether the file was malicious and assess if it had been previously reported within the cybersecurity community.
Additional notes	The file hash was submitted to VirusTotal for analysis, which revealed that multiple security vendors flagged it as malicious, classifying it as a Trojan (Qakbot). The community score was negative, confirming widespread consensus about the threat. The findings validated the initial alert and prompted escalation to the containment phase for further action on the affected workstation.

Need another journal entry template?

If you want to add more journal entries, please copy one of the tables above and paste it into the template to use for future entries.

Reflections/Notes: These activities helped me understand how different tools fit into real security work. Using tcpdump and Wireshark showed me how to capture and filter network traffic to find useful details. VirusTotal made it easy to confirm if a file was malicious, and the ransomware scenario showed how important prevention and quick response are. Overall, I learned how each tool and step connects to detecting and handling threats effectively.